National University of Computer and Emerging Sciences

Lab Manual 04

Programming Fundamentals

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Semester	Fall 2025

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Activities:

Activity 1:

Predict the behaviour(output) of the following programs. In case of any errors (syntax or logical) then suggest improvement.

1. `int main()`
`{`
`    int counter = 0;`
 	
`    while {counter > 100}`
`        if (counter % 2 == 1)`
`                cout << counter << " is odd" << endl;`
`            else`
`                cout << counter << " is odd" << endl;`
`            counter++;          // same as counter = counter + 1;`
 	
`    return 0;`
`}`

2. `int main()`
`{`
`    int K = 5;`
`    int I = -2;`
 	
`    while (I <= K)`
`    {`
`        I += 2;                 // same as I = I + 2`
`        --K;                    // same as K = K - 1`
`        cout << (I + K) << endl;`
`    }`
 	
`    return 0;`
`}`

3. `int main()`
`{`
`    int number = 4;`
 	
`    while (number >= 0)`
`        --number;               // same as number = number - 1;`
`    cout << number << endl;`
 	
`    return 0;`
`}`

Activity 2:

Write the programs using **while loop** for the following:

1. Print the following series:
    a. Print numbers from 1 to 100 with increment of 1.
    b. Print numbers from 100 to 1 with decrement of 1.
    c. Print numbers from 20 to 2 in steps of -2.
    d. Print sequence of numbers: 2, 5, 8, 11, 14, 17, 20.
    e. Print sequence of numbers: 99, 88, 77, 66, 55, 44, 33, 22, 11, 0.
2. Write a C++ program that uses a while loop to calculate and print the sum of the even integers from 2 to 30.
3. Write a C++ program that uses a while loop to calculate and print the product of the odd integers from 3 to 19.

4. Write a C++ program that uses while loops to find greatest common divisor (GCD) or highest common factor (HCF) of given two numbers (numbers can be positive and negative both).
5. Write a C++ program that print the square roots of the first 25 odd positive integers using while loops.
6. Write a C++ program that inputs a number from the user and displays its factorial using while loop. It asks the user whether user wants to calculate another factorial or not. If the user inputs 1, it again inputs number and calculate another factorial. If the user inputs 0, the program terminates.
7. Write a program in C++ that inputs a number from the user checks whether it is Armstrong number or not. If an n-digit number that is equal to the sum of its own digits, each raised to the power of n, the number is called an Armstrong number.
8. Because of the high price of petrol, you are concerned with the fuel consumption of your car. As a result, you have a record of the kilometres driven and litres used for each tank of petrol you purchase. Write a program that will display the kilometres driven, litres used, and consumption (in litres/100km) for each tankful. After processing all the input information, the program should calculate the overall average consumption:

   Enter the litres used (-1 to end): 57.6
   Enter the kilometres driven: 459
   The litres/100km for this tank was 12.5
   Enter the litres used (-1 to end): 45.3
   Enter the kilometres driven: 320
   The litres/100km for this tank was 14.2
   Enter the litres used (-1 to end): -1
   The overall average consumption was: 13.4